

Assignment: Object-Oriented Programming Fundamentals

Question 1 (30 points)

Explain the concept of encapsulation in Object-Oriented Programming.

Your answer must cover: (a) the definition of encapsulation, (b) how access modifiers implement it, (c) why encapsulation improves code maintainability.

Question 2 (40 points)

Explain the difference between a class and an object in OOP.

Your answer must cover: (a) what a class is (blueprint/template), (b) what an object is (instance), (c) the relationship between class and object, (d) provide a concrete example.

Question 3 (30 points)

Describe what inheritance is in OOP and explain one benefit of using it.

Your answer must include: (a) definition of inheritance, (b) the parent-child relationship, (c) one clear benefit such as code reuse.